

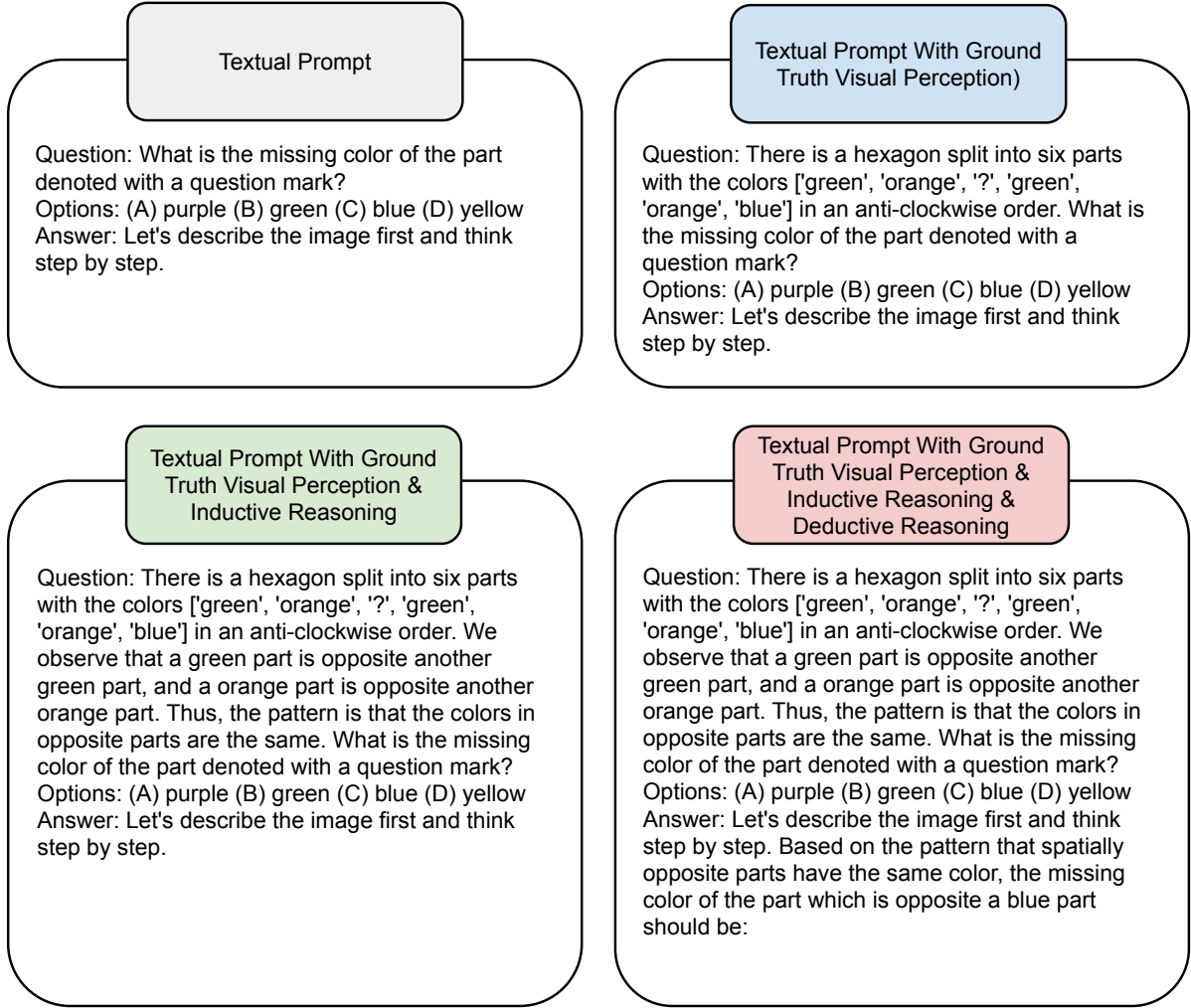


Figure 7: Textual prompt examples for eliciting reasoning steps from large multimodal models.

A.5 Dataset Size Analysis

Regarding the dataset size and diversity, we set the number of generated instances for each puzzle to 100, to reduce experimental variance and maintain a reasonable evaluation cost. Hence, the current dataset size is 2000 samples (20 templates with 100 instances each). As there are two templates per puzzle category, this means there are 200 test samples for each puzzle category. To observe the impact of the number of test samples, we evaluate the models on three different data settings as shown in Table 4: 50, 100, and 200 test samples in per puzzle, which correspond to 1000, 2000, and 4000 total samples respectively. In general, we observe some variations in the average score for single-concept puzzles, but it does not significantly affect the comparison of performance between different models. Hence, we believe that the chosen dataset size is large enough to be relatively robust to experimental variance. To investigate the multi-

	1000	2000	4000
Gemini Pro Avg. Score	29.2	34.5	32.2
GPT-4V Avg. Score	48.8	46.4	48.4

Table 4: Analysis of model performance with respect to number of testing data samples.

modal reasoning capabilities across diverse abstract scenarios, we construct the puzzles based on four fundamental concepts: numbers, colors, shapes, and size. Furthermore, to evaluate how well the models can relate to multiple concepts, we design both single-concept and dual-concept puzzles, and the taxonomy of diverse puzzles is shown in Figure 3.

	Numbers	Colors	Size	Shapes	Average
LLaVA-13B	25.5	29.0	37.0	39.0	32.6 (+5.1)
Gemini Pro	16.0	28.5	35.0	49.0	32.1 (-2.4)
GPT-4V	72.5	26.5	30.0	45.5	43.6 (-2.8)

Table 5: Results of direct prompting and change in average performance compared to CoT prompting.

A.6 Comparison of CoT and Direct Prompting

To investigate the effect of direct prompting without chain of thought, we evaluated the models on our main setting as shown in Table 1 and 2. The results are shown in Table 5 for single-concept puzzles and indicate that direct prompting is less effective for Gemini Pro and GPT-4V models, compared to CoT prompting in Table 1. This may be due to differences in training data and alignment methods between the models.

B Qualitative Analysis

To illustrate the reasoning bottlenecks of GPT-4V, we include two case study samples in Figure 8. For instance, the sample on the left is from the size & shapes category of puzzles, for which the model under-performed the random baseline. For visual perception, we observe that the model presents severe limitations, as it is unable to recognize simple polygon shapes and hallucinated additional shapes which are not in the image. Regarding inductive reasoning, we observe that the model was able to recognize the sizes of the different objects, but did not recognize the correct pattern that the circles directly adjacent to the center should be small in size. Hence, we believe that there is ample area for improvement for abstract reasoning ability in large multimodal models.

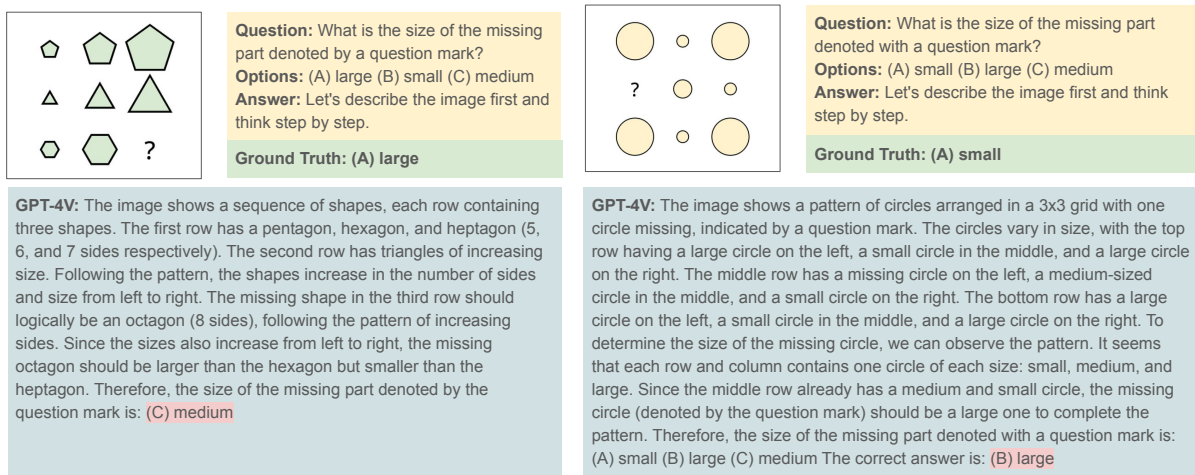


Figure 8: Case study on two sample predictions from GPT-4V. The example on the left shows visual perception failures and the example on the right shows the faulty inductive reasoning of the model which proposed a spurious pattern in the image.